information & broadcasting—have mounted an aggressive defence.

Its most vocal advocate remains Union road transport and highways minister Nitin Gadkari. "All this is false propaganda. There is no harm from ethanol," he says about the surge of complaints. Since taking office in 2014, Gadkari has championed ethanol-blended fuels as a way to reduce India's energy bill. Recently, he unveiled a flex-fuel version of the Maruti Suzuki WagonR and two such Hero MotoCorp motorcycles, compatible with ethanol blends ranging from E20 to E85. To support his claims about ethanol's safety, Gadkari points to extensive testing by the Automotive Research Association of India (ARAI). "After about 100,000 km of testing, ARAI has said that no vehicle engine failure was observed [by the use of ethanol-blended petrol]...no impact on vehicle performance, startability, drivability and metal capability," he says.

THE OFFICIAL RESEARCH

The government's case rests primarily on a couple of studies. A 2021 ARAI study examined how E20 affected eight metals, six rubber materials (elastomers) and four engineering plastics commonly used in vehicle fuel systems. It found "insignificant" corrosion in metals and broadly similar or, in some cases, better performance across other materials, although two elastomers and one plastic showed greater changes in some measured properties. The researchers said the findings would help refine fuel-system components.

The same year, an inter-ministerial expert committee constituted by NITI Aayog examined vehicle technology, ethanol supply and infrastructure before recommending a roadmap for a nationwide phased transition to E20 by 2025-26. It acknowledged three practical challenges: lower fuel efficiency in older engine calibrations, the need for E20-compatible materials in some fuel-system components, and the gradual introduction of E20-compatible vehicles alongside the fuel rollout.

Rather than carrying out fresh research, it relied on a 2014-15 study commissioned by the ministry of heavy industries that found no evidence of engine failures, drivability problems or abnormal engine wear from the use of E20. It did, however, identify greater degradation in a few elastomers and plastics, besides a 2-6 per cent decline in fuel efficiency. Addressing such concerns, ARAI director Reji Mathai recently acknowledged a similar decline with E20, describing it as "minimal". "Any variations in real-world fuel consumption and overall efficiency," he added, "are driven by external operational factors, including driver behaviour, road conditions and the baseline condition of the vehicle under test."

The National Policy on Biofuels, notified in 2018, targeted 20 per cent ethanol blending in petrol by 2030. When the target was advanced to 2025-26, the roadmap also envisaged achieving 10 per cent blending by November 2022. That milestone was reached in June 2022. From April 2023, all new petrol vehicles were required to be E20-compatible. "Obviously, after 2023, both the cars and the fuel are mandated for E20. That is not a cause of concern," says Rahul Bharti, executive director of Maruti Suzuki. "The concern is about cars manufactured and sold before 2023, which were primarily designed for E10." Bharti says Maruti began extensively testing E20 even before it became mandatory and found "no problem in terms of wear and tear, corrosion or damage" to either the life of the vehicle or the parts that come into contact with the fuel, including in pre-2023 models designed for E10.

THE E20 DEBATE

E20 is a blend of 20 per cent ethanol—a renewable biofuel produced by fermenting crops such as sugarcane and maize—and 80 per cent petrol. Introduced as the default petrol across India from April, it has triggered several claims about its impact on vehicles



CLAIM: E20 cuts mileage by 30%

GOVT RESPONSE: This figure refers to ethanol's lower calorific value, not real-world mileage, which depends on driving habits, tyre pressure, AC load etc.



CLAIM: E20 damages engines, especially older ones

GOVT RESPONSE: Blend cleared after extensive testing; no widespread pattern of engine failure



CLAIM: Ethanol reduces vehicle performance

GOVT RESPONSE: Ethanol is a high-octane fuel. E20 can improve combustion in calibrated vehicles



CLAIM: Blended fuel should be cheaper

GOVT RESPONSE: Ethanol was cheaper in 2020-21, but procurement costs have since risen. The programme continues because of energy security, lower emissions and farm-income benefits



CLAIM: Insurance claims/ warranties will be denied due to E20 use

GOVT RESPONSE: Insurers and original equipment manufacturers deny the claim; automakers' body SIAM says warranties will be honoured for spec-compliant E20



CLAIM: Sugarcane juice is mixed directly with petrol

GOVT RESPONSE: Ethanol is produced through fermentation and industrial processing

Source: Ministry of Petroleum & Natural Gas